Private 5G Network Setup Guide

Open5GS Core + srsRAN gNB + USRP X310

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1 Overview

This document details the configuration required to deploy a standalone 5G network using **Open5GS** (Core Network) and **srsRAN Project** (gNodeB) with a **USRP X310** radio. The setup utilizes Linux Network Namespaces to isolate the Radio Access Network (RAN) from the Core, simulating a real-world remote deployment.

2 Prerequisites

- **OS:** Ubuntu 22.04 LTS (Real-time kernel recommended)
- **Hardware:** USRP X310 (connected via Ethernet to `eno1`), 5G UE (Samsung Smartphone), Programmable USIM.
- **Software:** Open5GS, srsRAN Project, UHD 4.x Drivers.

3 Step 1: Network Namespace Architecture

To prevent IP conflicts and simulate network separation, the gNodeB runs inside a namespace called `ueran-ns`.

3.1 Namespace Setup Script (`setup_ns.sh`)

Create the virtual environment and network cables (veth pairs).

```
#!/bin/bash
# 1. Create Namespace
sudo ip netns add ueran-ns

# 2. Create Virtual Cable (veth pair)
sudo ip link add veth-core type veth peer name veth-ran
sudo ip link set veth-ran netns ueran-ns

# 3. Configure Host Side (Core Network Interface)
sudo ip addr add 10.200.1.1/24 dev veth-core
sudo ip link set veth-core up

# 4. Configure Namespace Side (gNodeB Interface)
sudo ip netns exec ueran-ns ip addr add 10.200.1.2/24 dev veth-ran
sudo ip netns exec ueran-ns ip link set veth-ran up
sudo ip netns exec ueran-ns ip link set lo up

# 5. Route & NAT (Enable Internet Access for UE)
sudo ip netns exec ueran-ns ip route add default via 10.200.1.1
sudo sysctl -w net.ipv4.ip_forward=1
sudo iptables -t nat -A POSTROUTING -s 10.200.1.0/24 -j MASQUERADE
sudo iptables -t nat -A POSTROUTING -s 10.45.0.0/16 -j MASQUERADE
```

4 Step 2: USRP Hardware Isolation

A critical step is moving the physical Ethernet interface (`eno1`) into the namespace so the gNodeB can see the radio.

4.1 Procedure

Run the following commands **after** running the namespace setup script:

```
# Identify your USRP interface (e.g., eno1)
# Move physical interface into the namespace
sudo ip link set eno1 netns ueran-ns

# Assign IP to the USRP Host Interface inside the namespace
# (Assuming USRP IP is 192.168.10.2, Host must be 192.168.10.1)
sudo ip netns exec ueran-ns ip addr add 192.168.10.1/24 dev eno1
sudo ip netns exec ueran-ns ip link set eno1 up

# Set MTU for performance (Jumbo Frames)
sudo ip netns exec ueran-ns ip link set eno1 mtu 9000
```

4.2 Firmware Setup (If Version Mismatch Occurs)

If `uhd_find_devices` fails due to FPGA incompatibility, update the firmware *inside* the namespace:

```
# 1. Download images on Host
sudo uhd_images_downloader

# 2. Burn image inside Namespace
sudo ip netns exec ueran-ns uhd_image_loader --args="type=x300,addr
    =192.168.10.2"
```

5 Step 3: Core Network Configuration (Open5GS)

Ensure the AMF listens on the IP address accessible by the namespace (`10.200.1.1`).

5.1 File: `amf.yaml`

Location: `install/etc/open5gs/amf.yaml`

```
amf:
  ngap:
    server:
      - address: 10.200.1.1    # Must match veth-core IP
  guami:
    - plmn_id:
        mcc: 001
        mnc: 01                # Must match SIM card
  tai:
    - plmn_id:
        mcc: 001
        mnc: 01
      tac: 1                   # Must match gNodeB Config
```

6 Step 4: gNodeB Configuration (srsRAN)

The configuration file must be placed in `/opt` to avoid AppArmor permission issues in the home directory.

6.1 File: /opt/srsRAN_gnb/gnb_config.yml

```
cu_cp:
  amf:
    addr: 10.200.1.1          # IP of Core (Host)
    bind_addr: 10.200.1.2     # IP of gNodeB (Namespace)
  supported_tracking_areas:
    - tac: 1                  # Must match AMF
      plmn_list:
        - plmn: "00101"

ru_sdr:
  device_driver: uhd
  # Explicitly point to USRP IP
  device_args: type=x300,addr=192.168.10.2,num_recv_frames=64,num_send_frames
     =64
  # Use sc16 to avoid compression errors
  otw_format: sc16
  # Gains adjusted for X310 limits
  tx_gain: 30                 # Max ~31.5
  rx_gain: 30                 # Max ~37.5

cell_cfg:
  dl_arfcn: 632628            # Band n78 (3.5 GHz)
  band: 78
  channel_bandwidth_MHz: 20
  common_scs: 30
  plmn: "00101"
  tac: 1
```

7 Step 5: Execution Sequence

To start the network from a fresh reboot, execute in this order:

1. **Setup Namespace:** `./setup_ns.sh`
2. **Move USRP Interface:** Execute commands from Section 2.1.
3. **Start Core:** `./start_5g.sh`
4. **Start gNodeB:**

```
sudo ip netns exec ueran-ns /opt/srsRAN_gnb/gnb -c /opt/srsRAN_gnb/
    gnb_config.yml
```

8 Troubleshooting Reference

- **Permission Denied:** Move binary and config to `/opt` to bypass AppArmor/Home directory restrictions.
- **Bind failed (Cannot assign requested address):** Check `amf.addr` vs `bind_addr`. Ensure the IP exists in `ip addr`.
- **No Devices Found:** The USRP interface (`eno1`) is not inside the namespace. Use `ip link set eno1 netns ueran-ns`.
- **NG Setup Failed (Unknown PLMN/SNPN):** The TAC in `gnb_config.yml` (e.g., 7) does not match the TAC in `amf.yaml` (e.g., 1).
- **WebUI Address In Use:** An old process is running. Use `sudo lsof -i :9999` to find PID and `kill -9 PID`.